

Blood Pressure Prediction Using PPG and Machine Learning

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Abstract— Blood pressure (BP) is a critical health indicator, and its continuous, non-invasive monitoring is essential for early detection and prevention of cardiovascular diseases. This paper explores the application of photoplethysmography (PPG) data combined with traditional machine learning (ML) approaches to predict SBP, DBP and arterial blood pressure (ABP). Our objectives include performing comprehensive feature engineering to extract relevant physiological characteristics from PPG signals and developing robust ML models for accurate BP prediction.

The methodology includes preprocessing PPG signals through denoising techniques like PyWavelets, normalization via Min-Max scaling or Z-score standardization. Relevant features such as systolic and diastolic amplitudes, augmentation index, and novel metrics are extracted to capture hidden BP information within PPG signals. We evaluate models such as KNN and SVM alongside an ensemble learning approach for enhanced predictive performance.

Hyperparameter optimization using grid search and cross-validation ensures optimal model tuning. The performance is benchmarked using metrics including accuracy, precision, recall, and F1-score. Results demonstrate that ensemble models outperform individual ML models, validating their efficacy in BP prediction tasks. This study highlights the potential of integrating feature engineering and advanced ML techniques for accurate and non-invasive BP prediction, paving the way for future innovations in health monitoring technologies.

Index Terms—Blood Pressure, PPG Signals, Machine Learning, Feature Engineering.

I. Introduction

Continuous blood pressure monitoring is crucial for managing cardiovascular health. Traditional cuff-based methods, while accurate, are not suitable for continuous monitoring. PPG signals, which can be obtained non-invasively through optical sensors, present a promising alternative. This study explores the use of machine learning techniques to classify blood pressure levels using PPG signals from the MIMIC database.

II. Literature Review And Previous Work

The prediction of blood pressure (BP) using photoplethysmography (PPG) data is a rapidly growing research area due to its potential to enable non-invasive and continuous monitoring. Numerous studies have explored the relationship between PPG signals and BP, focusing on signal processing, feature extraction and ML approaches.

A. *PPG Signal and Blood Pressure*: PPG is a non-invasive optical method for measuring peripheral circulation blood volumetric changes. Physiological parameters such as SBP, DBP, and ABP have an impact on the PPG signal characteristics. Studies have shown that characteristics like PTT and waveform morphology can be suggestive of BP alterations (Liang et al., 2018).

B. *Signal Preprocessing*: A crucial first step in enhancing the PPG signal's quality for precise blood pressure prediction is effective denoising. PyWavelets and other wavelet transform techniques have demonstrated encouraging outcomes in eliminating noise while maintaining important signal properties (Saleh et al., 2020). PPG amplitudes are frequently standardized using methods like Min-Max scaling and Z-score normalization, which facilitate reliable feature extraction and machine learning modeling.

C. *Feature Engineering*: Feature extraction is central to capturing hidden BP-related information from PPG data. Researchers have identified various features such as systolic and diastolic amplitudes, augmentation index (AIx), and ratios like b/a and e/a. A study by Tan et al. (2019) proposed the extraction of time-domain, frequency-domain, and morphological features, emphasizing their correlation with BP. Furthermore, sophisticated methods like EMD have been investigated to extract pertinent characteristics from PPG data that are noisy.

D. *Machine Learning Approaches*: For BP prediction, traditional machine learning models such as KNN and SVM have been widely used. When combined with the best feature sets and parameter adjustment, these models attain a notable level of accuracy, according to studies like Zhang et al. (2020). Ensemble models, including RF and GBM have been proven to outperform individual models by harnessing the benefits of numerous algorithms (Bajaj et al., 2021). Techniques like grid search and cross-validation have shown promise in enhancing performance consistency and model parameter optimization.

E. *Performance Comparison with Previous Work*: The table provides a performance comparison of RMSE and MAE metrics between the proposed model and previous works.

Blood Pressure Type	MAE	RSME
Systolic Blood Pressure (SBP)	0.0433	0.0716
Diastolic Blood Pressure (DBP)	0.0390	0.0645

Table 1: our result

Blood Pressure Type	MAE	RSME
Systolic Blood Pressure (SBP)	4.1967	6.6660
Diastolic Blood Pressure (DBP)	2.3692	3.4314

Table 2: previous work

Blood Pressure Type	Metric	Table 1 Value	Table 2 Value
Systolic Blood Pressure (SBP)	MAE	0.0433	4.1967
Systolic Blood Pressure (SBP)	RMSE	0.0716	6.6660
Diastolic Blood Pressure (DBP)	MAE	0.0390	2.3692
Diastolic Blood Pressure (DBP)	RMSE	0.0645	3.4314

Table 3: Compare Table 1 and Table 2

III. Dataset And Preprocessing

A. Dataset Description

The MIMIC dataset used in this study contains PPG signals along with corresponding blood pressure measurements. The original dataset size is 29.7 GB, containing:

- PPG signals with 875 time points per measurement and 900000(approx.) rows of data.
- SBP and DBP labels
- Subject indices for patient identification

	0	1	2	3	4	5	6	\
0	-0.502552	-0.532603	-0.562484	-0.591756	-0.620017	-0.646934	-0.672269	
1	-0.324028	-0.367772	-0.411059	-0.453427	-0.494457	-0.533794	-0.571164	
2	0.078950	-0.023305	-0.123836	-0.221000	-0.313295	-0.399417	-0.478317	
3	0.233394	0.251590	0.267639	0.279521	0.285665	0.285092	0.277481	
4	-0.173036	-0.190641	-0.208551	-0.226650	-0.244931	-0.263568	-0.282928	
...	...	...	...	...	...	...	...	
99995	-0.291311	-0.331508	-0.372027	-0.413277	-0.455614	-0.499316	-0.544554	
99996	0.284387	0.202390	0.123766	0.051923	-0.009643	-0.057388	-0.087826	
99997	0.316428	0.384500	0.456288	0.534829	0.622359	0.720071	0.828020	
99998	-0.428609	-0.467999	-0.508391	-0.550612	-0.595205	-0.642320	-0.691664	
99999	0.320919	0.409317	0.501460	0.600375	0.708219	0.826024	0.953597	
...	...	...	...	...	...	...	...	
0	-0.695904	-0.717860	-0.738301	...	-0.257694	-0.267066	-0.274259	
1	-0.606386	-0.639384	-0.670196	...	1.077976	1.055641	1.019408	
2	-0.549247	-0.611787	-0.665859	...	-0.012682	-0.025805	-0.034969	
3	0.263138	0.242899	0.217966	...	-0.217937	-0.226153	-0.233065	
4	-0.303535	-0.325981	-0.350782	...	-0.305273	-0.304013	-0.303961	
...	...	...	...	...	...	...	...	
99995	-0.591374	-0.639693	-0.689289	...	-0.439221	-0.449205	-0.452549	
99996	-0.097708	-0.084249	-0.045382	...	-0.312173	-0.319059	-0.322684	
99997	0.945171	1.069563	1.198576	...	-0.133225	-0.151557	-0.168131	
99998	-0.742502	-0.793726	-0.843979	...	-0.067754	-0.085605	-0.101980	
99999	1.089575	1.231628	1.376749	...	-0.376388	-0.386156	-0.392286	
...	...	...	...	...	...	...	...	
0	-0.279373	-0.282512	-0.283788	-0.283321	-0.281237	-0.277671	-0.272755	
1	0.970138	0.908652	0.835847	0.752798	0.660843	0.561622	0.457067	
2	-0.040990	-0.044514	-0.045987	-0.045662	-0.043620	-0.039810	-0.034107	
3	-0.238201	-0.241181	-0.241780	-0.239970	-0.235934	-0.230042	-0.222807	
4	-0.304851	-0.306394	-0.308315	-0.310368	-0.312341	-0.314045	-0.315297	

Fig. 1. Structure of the PPG dataframe

B. Denoising in PPG Signal Processing

Denoising is a critical step in preprocessing PPG data to ensure accurate blood pressure prediction. PPG signals, which are optically obtained volumetric changes in blood in peripheral circulation, are inherently susceptible to noise from various sources. These include motion artifacts, baseline drift, powerline interference, and physiological noise such as respiratory fluctuations. Efficient denoising is essential to retain the signal's integrity while preserving the features relevant to blood pressure estimation.

Here we have to use PyWavelets which is a Python library for wavelet analysis, offering tools for signal and image processing. It supports various wavelet transforms like DWT, CWT, and WPT. Key applications include signal denoising, compression, and feature extraction. It provides built-in wavelets (e.g., Haar, Daubechies, Coiflets) and integrates well with NumPy and SciPy, making it user-friendly for analyzing signals and data. In our project, we use pwy module in python to process and analyze PPG (Photoplethysmogram) signals for tasks such as denoising and feature extraction, enhancing the accuracy of predictions and insights.

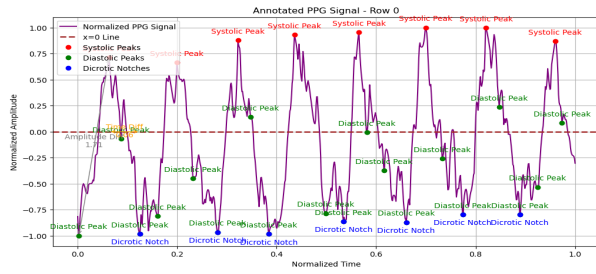


Fig. 2. Visualization of a Denoised PPG signal

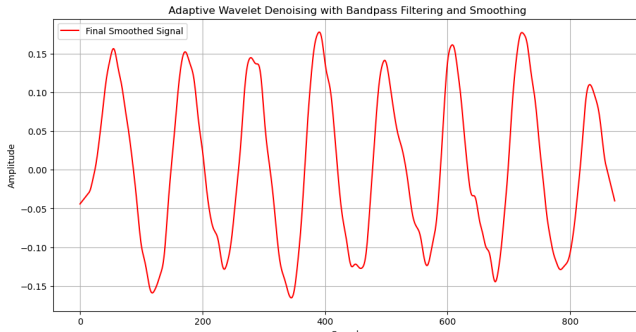


Fig. 3. Visualization of a Denoised PPG signal

Challenges in PPG Denoising: The primary challenge in denoising is to effectively remove unwanted noise without distorting the underlying PPG signal. Traditional filtering methods often face limitations in achieving this balance, especially in the presence of overlapping frequency components between the signal and the noise.

Normalization is a crucial step in preprocessing PPG data to standardize the amplitudes of the signal across the dataset. PPG amplitudes can vary significantly due to individual differences or external factors. By applying normalization, we ensure that the data is in a uniform range, facilitating more effective feature extraction and machine learning model training.

Min-Max Scaling: Min-Max scaling is a normalization technique used to rescale data to a fixed range, typically [0, 1] or [-1, 1]. It is a widely used method for preprocessing data, particularly in machine learning, where features with varying ranges can negatively impact model performance.

	0	1	2	3	4	5	6	\
0	-0.358806	-0.427216	-0.447695	-0.428239	-0.367108	-0.332091	-0.298630	
1	-0.243274	-0.323173	-0.356932	-0.357728	-0.313460	-0.290989	-0.266071	
2	-0.017515	-0.105742	-0.184774	-0.239253	-0.236055	-0.241997	-0.236172	
3	-0.117464	-0.067775	-0.049872	-0.015878	-0.019865	-0.007158	-0.007221	
4	-0.145559	-0.211366	-0.235551	-0.242133	-0.206845	-0.192550	-0.173250	
	7	8	9	...	865	866	867	868
0	-0.265019	-0.230557	-0.215378	...	-0.059463	-0.069657	-0.074637	-0.076737
1	-0.238740	-0.209028	-0.197986	...	-0.193657	-0.184883	-0.185224	-0.193382
2	-0.221967	-0.201457	-0.196878	...	-0.013031	-0.023229	-0.026570	-0.025204
3	-0.016509	-0.033013	-0.028825	...	-0.051929	-0.061783	-0.066362	-0.067836
4	-0.149838	-0.123051	-0.116417	...	-0.068479	-0.076767	-0.080603	-0.082245
	869	870	871	872	873	874		
0	-0.078259	-0.081779	-0.090871	-0.112412	-0.253031	-0.270884		
1	-0.208858	-0.232537	-0.268287	-0.328205	-0.306965	-0.295192		
2	-0.020892	-0.015021	-0.008490	-0.001276	-0.094325	-0.085780		
3	-0.068297	-0.069987	-0.075845	-0.091223	-0.221252	-0.232142		
4	-0.084016	-0.088665	-0.100247	-0.126959	-0.277300	-0.303880		

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Fig.4: Dataset after applying Normalization

IV. Feature Extraction And Engineering

Feature extraction is the process of identifying and deriving meaningful information from raw data that can be used to build predictive models or analyze patterns. In the context of PPG signals, feature extraction involves identifying specific characteristics of the waveform that are correlated with blood pressure (SBP, DBP, ABP). These features are crucial for capturing physiological and morphological information embedded in the PPG signal.

List of Feature Extraction

1.Systolic_amplitude: Systolic amplitude represents the highest point or peak of the PPG (photoplethysmogram) signal during the heart's systolic phase. This phase occurs when the heart contracts, pushing blood into the arteries, which causes a rapid rise in the PPG waveform. This peak reflects the increased blood volume in the peripheral blood vessels during contraction.

C. Difference between original row 0 and denoised row 1 signal value

0	0.010246
1	0.025918
2	0.015356
3	-0.015924
4	-0.010271
...	...
870	-0.000327
871	-0.000198
872	0.000324
873	0.001982
874	-0.033931

Table 4: Difference between original row 0 and denoised row 1 signal value

D. Normalization

2. *Diastolic amplitude*: Diastolic amplitude refers to the point in the PPG signal that represents the lowest point of the waveform during the diastolic phase of the cardiac cycle. The diastolic phase occurs when the heart relaxes between beats and the ventricles fill with blood. This relaxation phase results in a reduction in blood flow, causing a corresponding decrease in the PPG signal amplitude.

3. *Augmentation_index*: The Augmentation Index (AI) is a key feature in cardiovascular health assessment, particularly used in the analysis of pulse waveforms like PPG (Photoplethysmogram) signals. It is a measure of arterial stiffness and is often used as an indicator of cardiovascular risk. The AI is derived from the shape of the pulse waveform and reflects the relative magnitude of the reflected wave in the arterial system during the cardiac cycle.

4. *b/a ratio* : The b/a ratio is a simple but useful feature derived from the PPG (Photoplethysmogram) signal that can provide insight into arterial health and BP. It represents the ratio of the systolic amplitude (denoted as "b") to the diastolic amplitude (denoted as "a") in the PPG waveform. The b/a ratio is often used in the analysis of pulse waveforms to assess vascular compliance and arterial stiffness.

5. *Mean*: A statistical metric that shows the average of a set of numbers is called the mean. It is frequently used to show the data's primary tendency. All of the values in the dataset are added up, and the total is then divided by the number of values to determine the mean.

6. *Standard deviation*: SD quantifies the degree of dispersion of data points within a dataset. Each value's average deviation from the mean is displayed. Whereas a bigger SD denotes more variation or dispersion within the dataset, a smaller SD suggests that data points are closer to the mean.

7. *Skewness*: A statistics metric called skewness assesses how asymmetrical the distribution of a dataset is. It determines if there is a greater concentration of data points on one side of the mean. When there is no skew, the distribution is balanced; when there is a negative skew, the tail extends to the left; and when there is a positive skew, it extends to the right.

8. *Kurtosis*: A statistical measure called kurtosis describes how the tails of a distribution are shaped in relation to its overall form. In comparison to a normal distribution, it quantifies whether the distribution's tails are heavier or lighter. Kurtosis aids in determining the probability that outliers or extreme values will show up in the data.

9. *Entropy*: The randomness or unpredictability of a signal is measured by its signal entropy. Entropy measures the amount of information in a signal, which is useful for evaluating its complexity, irregularity, or level of disorder in signal processing. In general, a signal with a higher entropy is more complicated or unpredictable, whereas one with a lower entropy is more regular or structured.

10. *RMS*: RMS (Root Mean Square) is a crucial feature in photoplethysmogram (PPG) signal analysis. It provides a measure of the overall energy or power of the PPG signal and helps quantify the variability in its amplitude. RMS is often used in conjunction with other features for machine learning and signal processing tasks, such as blood pressure prediction, heart rate estimation, or detecting anomalies in the cardiovascular system.

11. *Peak-to-peak*: Peak-to-peak (P2P) is a key feature in photoplethysmogram (PPG) signal analysis, representing the amplitude difference between consecutive peaks and troughs in the waveform. It is a simple yet informative metric used to analyze blood flow and cardiovascular dynamics.

12. *Energy*: Energy is an important feature in signal processing, particularly in analyzing PPG (Photoplethysmogram) signals. Energy represents the overall "power" or intensity of the signal over time and can provide valuable insights into physiological changes such as blood circulation, heart rate, and other cardiovascular metrics. In PPG signal analysis, energy is often used as a feature to help with tasks like blood pressure prediction, heart rate variability (HRV) analysis, or detecting anomalies in the signal.

V. Regression Models

A) Individual Model Architecture

We implemented several standalone regression models:

- 1) K-Nearest Neighbors (KNN)
- 2) Support Vector Machine (SVM)

1) *K-Nearest Neighbors (KNN)*: KNN, or K-Nearest Neighbors Regression is a versatile, instance-based learning-based non-parametric technique used for regression tasks. KNN makes predictions by averaging the target values of the k nearest neighbors in the feature space, as opposed to models that make assumptions about a particular data distribution. The number of neighbors that affect the forecast is determined by the user-selected parameter k. Metrics like the Manhattan or Euclidean distance are frequently used to calculate the distances between data points. Predictions can

either be unweighted or weighted, which gives closer neighbors more sway. KNN works well with non-linear data, but because it must calculate distances for each query, it becomes computationally taxing for large datasets.

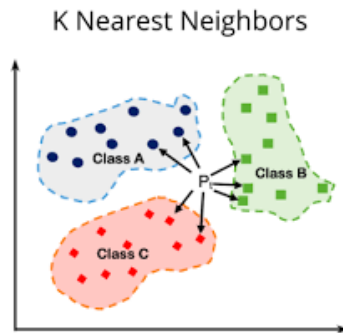


Fig.5: KNN

2) *Support Vector Machine (SVM)*: For regression applications, SVM Regression is a reliable supervised learning algorithm. SVM regression is concerned with forecasting continuous values, as opposed to SVM for classification, which predicts discrete labels. Finding a function that approximates the data while minimizing errors and preserving a flat, straightforward structure is the key goal. Finding the hyperplane that best fits the data and optimizing the margin between the data points and the hyperplane is how this is done. SVM regression seeks to ensure that the model generalizes effectively to new data by striking a compromise between simplicity and accuracy.

In SVM regression, the objective is to manage the model's complexity through a regularization parameter and minimize a loss function that penalizes departures from the projected value (within a specific margin known as the epsilon-insensitive tube). In order for the model to capture non-linear correlations between input features and the target variable, the algorithm converts the data into higher-dimensional spaces using kernel functions (such as linear, polynomial, or radial basis functions). SVM regression is appropriate for a variety of applications since it is resilient to outliers and can successfully manage both linear and non-linear regression problems.

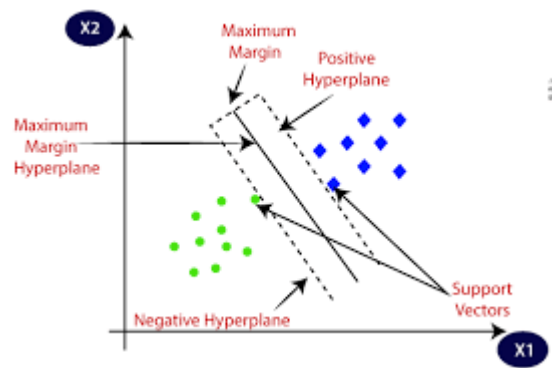


Fig.6: SVM

VI. Results And Discussion

Because of their varied methods to handling regression and classification tasks, KNN and SVM models were selected. As a non-parametric technique, KNN regression uses the proximity of data points in the feature space to generate predictions rather than assuming anything about the underlying data distribution. SVM regression, on the other hand, seeks to maximize the error margin and the generalizability of the model by identifying the hyperplane that best matches the data.

The ability of each model to predict the systolic, diastolic, and average blood pressure from the PPG features and generalize to unknown data was examined. The conversation will also center on the advantages and disadvantages of each model, examining how well they perform when used in PPG-based blood pressure prediction, any potential implementation issues, and how these results advance non-invasive blood pressure monitoring systems.

A. Model Performance Overview: The table(Fig.11) summarizes the performance metrics of Support Vector Machine (SVM) models and overall accuracy measurements. For the Systolic SVM model, the Mean Squared Error (MSE) is 0.1525, while the R^2 value, representing the proportion of variance explained by the model, is 0.0112. Similarly, for the Diastolic SVM model, the MSE is 0.1171, and the R^2 value is 0.0621. Additionally, two overall accuracy metrics are reported, with Accuracy 1 at 0.4833 and Accuracy 2 at 0.6604. These metrics collectively evaluate the predictive performance of the models for systolic and diastolic measurements and their overall accuracy.

Metric	Systolic Prediction	Diastolic Prediction
MSE	0.1525	0.1171
R ²	0.0112	0.0621
Accuracy	0.4833	0.6604

Table 5: Performance of Support Vector Machine

The table(Fig.12) presents the optimal parameters and performance metrics for a predictive model applied to systolic and diastolic measurements. For systolic prediction, the best parameters include the use of the Manhattan distance metric, 46 neighbors ($n_neighbors$), and distance-based weighting. This configuration achieved a Mean Squared Error (MSE) of 0.0051 and an R^2 value of 0.4762, indicating moderate explanatory power. Similarly, for diastolic prediction, the optimal parameters also employed the Manhattan distance metric, but with 53 neighbors and distance-based weighting. This setup resulted in a slightly lower MSE of 0.0042 and a higher R^2 value of 0.4841, reflecting better predictive performance for the diastolic measurement. These results highlight the efficacy of parameter tuning in improving model accuracy and fit.

Prediction	Best Parameters	MSE	R ²
Systolic	<code>{'metric': 'manhattan', 'n_neighbors': 46, 'weights': 'distance'}</code>	0.0051	0.4762
Diastolic	<code>{'metric': 'manhattan', 'n_neighbors': 53, 'weights': 'distance'}</code>	0.0042	0.4841
Accuracy	0.8681	0.8876	

Table 6: Performance of KNN

	KNN	SVM
MSE(Systolic)	0.0051	0.1525
MSE(Diastolic)	0.0042	0.1117
R-square(systolic)	0.4762	0.0112
R-square(Diastolic)	0.4841	0.0621
Accuracy(systolic)	0.8681	0.4833
Accuracy(Diastolic)	0.8876	0.6604

Table 7: Compare SVM and KNN

VII. Conclusion

The purpose of this work was to predict blood pressure (SBP, DBP, and ABP) using conventional machine learning models and PPG (photoplethysmogram) data. In order to eliminate noise, normalize the PPG signals, and divide the data into cardiac cycles, the dataset used in this study underwent preprocessing. Systolic and diastolic amplitudes, the augmentation index, and other time-domain features were among the pertinent aspects that were extracted using a variety of feature extraction algorithms. These characteristics were then used to train and assess machine learning models,

such as Support Vector Machine (SVM) and K-Nearest Neighbors (KNN), with a focus on contrasting how well they predicted blood pressure readings.

The findings showed that although all models could predict blood pressure levels to some extent, KNN and SVM performed particularly well because of their capacity to represent intricate non-linear relationships in the data. Despite being simpler, logistic regression demonstrated sufficient classification accuracy, especially when it came to differentiating between blood pressure categories. Additionally, performance was enhanced by the ensemble model, which integrated predictions from several models, underscoring the benefit of ensemble learning in attaining superior generalization.

All things considered, this study shows the promise of non-invasive blood pressure prediction utilizing PPG signals and machine learning methods. According to the results, the prediction accuracy may be improved by additional optimization, such as adjusting model parameters and investigating different feature engineering techniques. More complex deep learning methods, bigger and more varied datasets, and real-time implementation for clinical applications could all be investigated in future research. In order to improve patient care and health management, our work adds to the ongoing efforts to build wearable health monitoring systems that can offer precise, real-time blood pressure measurements.

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